Scalar Field Dark Energy Model

Data Analysis Techniques

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Outline

1. Statistics
2. The method of maximum likelihood - likelihood function
3. The method of least squares - χ^2 function
4. χ^2 function of ϕCDM moedel
5. Likelihood function of ϕCDM moedel
6. Nuisance parameters
7. Constraints on individual cosmological parameters

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- For random variable x, given a *probability density function* (p.d.f.), $f(x;\theta)$, which may depend on one or more parameters θ.
- The *expectation value* of any function $u(x)$ is

$$E[u(x)] = \int_{-\infty}^{\infty} u(x) f(x;\theta) dx \ .$$

- The nth moment of a random variable is

$$\alpha_n \equiv E[x^n] = \int_{-\infty}^{\infty} x^n f(x;\theta) dx \ .$$

- The nth central moment of x is

$$m_n \equiv E[(x-\alpha_1)^n] = \int_{-\infty}^{\infty} (x-\alpha_1)^n f(x;\theta) dx \ .$$

- The most commonly used moments are the mean μ and variance σ^2:

$$\mu \equiv \alpha_1 \ ,$$

$$\sigma^2 \equiv V[x] \equiv m_2 = \alpha_2 - \mu^2 \ .$$

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- A set of N measure quantities $\boldsymbol{x} = (x_1, ..., x_N)$ describe by a joint p.d.f. $f(\boldsymbol{x}; \boldsymbol{\theta})$, where $\boldsymbol{\theta} = (\theta_1, ..., \theta_n)$ is a set of n parameters whose value are unknown.
- The *likelihood function* $\mathcal{L}(\boldsymbol{\theta}) \equiv f(\boldsymbol{x}; \boldsymbol{\theta})$.
- If the measurements x_i are statistically independent and each follow the p.d.f. $f(x_i; \boldsymbol{\theta})$, then the joint p.d.f. for x factorizes and the likelihood function is

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^{N} f(x_i; \boldsymbol{\theta}) \ .$$

- It is usually easier to work with $\ln \mathcal{L}$, and since both are maximized for the same parameter value $\boldsymbol{\theta}$, the maximum likelihood (ML) estimators can be found by solving the *likelihood equations*,

$$\frac{\partial \ln \mathcal{L}}{\partial \theta_i} = 0 \ , \qquad i = 1, ..., n \ .$$

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- The *method of least squares* (LS) coincides with the method of maximum likelihood in the following special case.
- Consider a set of N independent measurements y_i at known points x_i. The measurement y_i is assumed to be Gaussian distributed with mean $F(x_i;\boldsymbol{\theta})$ and known variance σ_i^2.
- The goal is to construct estimators for the unknown parameters $\boldsymbol{\theta}$,

$$\chi^2(\boldsymbol{\theta}) = -2\ln\mathcal{L}(\boldsymbol{\theta}) + \text{constant} = \sum_{i=1}^{N} \frac{(y_i - F(x_i;\boldsymbol{\theta}))^2}{\sigma_i^2} \ .$$

- The set of parameters $\boldsymbol{\theta}$ which maximize $\mathcal{L}$ is the same as those which minimize χ^2.

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- The Lagrangian density for ϕCDM moedel

$$\mathcal{L} = -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \ .$$

- Given the specific potential

$$V(\phi) = \frac{\kappa}{2G} \phi^{-\alpha} \ ,$$

$$\Rightarrow \begin{cases} \rho_\phi = -T_{00}^{(\phi)} = \frac{1}{2}\dot{\phi}^2 + V(\phi) \\ P_\phi = T_{ii}^{(\phi)} = \frac{1}{2}\dot{\phi}^2 - V(\phi) \end{cases} \ .$$

- We can take the Hubble parameter as the χ^2 function or likelihood function

$$H(\alpha, \Omega_{\rm m}) = H_0 \sqrt{\Omega_{\rm m} + \Omega_\phi(\alpha)} \ .$$

- N measurements X_i^{obs} at redshifts z_i with measurements errors σ_i, and the ϕCDM model predicts $X_i^{th}(\alpha, \Omega_{\rm m})$.
- The χ^2 function is

$$\chi^2(\alpha, \Omega_{\rm m}) = \sum_i \frac{[X^{obs}(z_i) - X^{th}(z_i, \alpha, \Omega_{\rm m})]^2}{\sigma_i^2} \ .$$

- Small values of χ^2 indicate a good fit. The parameters α^* and Ω^* that minimize χ^2 are called the best-fit parameters.

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- The likelihood function

$$\begin{aligned}\mathcal{L}(\alpha, \Omega_{\rm m}) &= \exp\left(\frac{-\chi^2(\alpha, \Omega_{\rm m})}{2}\right) \\ &= \prod_i \exp\left[-\frac{[X^{obs}(z_i) - X^{th}(z_i, \alpha, \Omega_{\rm m})]^2}{2\sigma_i^2}\right],\end{aligned}$$

  has a lacal maximum $\mathcal{L}_{\rm ML}$ at $(\alpha^*, \Omega_{\rm m}^*)$.
- 1,2, and 3σ confidence level contours are defined as the pairs of parameters $(\widehat{\alpha}, \widehat{\Omega_{\rm m}})$ for which $-2\ln\mathcal{L}(\widehat{\alpha}, \widehat{\Omega_{\rm m}})$ is less by 2.30, 6.18, and 11.83 respectively compared to $\mathcal{L}_{\rm ML}$.

- If the likelihood function had an exact two dimensional Gaussian profile,

$$\mathcal{L} \propto e^{-c_1(\alpha-\alpha^*)^2+(\Omega_{\rm m}-\Omega_{\rm m}^*)^2/(2\sigma^2)}$$

- $N\sigma$ contours would correspond to the range of parameters $(\alpha^* \pm N\sigma, \Omega_{\rm m}^* \pm N\sigma)$.

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- These "nuisance" parameters $\boldsymbol{\nu}$ can be determined from the data but are of no interest to us.
- The information of $\boldsymbol{\nu}$ can be summarize in the form of probability density function $P(\boldsymbol{\nu})$.
- $P(\boldsymbol{\nu})$ is normalized to one and is usually taken to be a Gaussian peaked at the best-fit value of $\boldsymbol{\nu}^*$ determined from other independent data set.
- We will determined the best-fit values and confidence level contours from $\mathcal{L}'$,

$$\mathcal{L}'(\alpha, \Omega_{\rm m}) = \int \mathcal{L}(\alpha, \Omega_{\rm m}, \boldsymbol{\nu}) P(\boldsymbol{\nu}) d\boldsymbol{\nu} \ .$$

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- One-dimensional confidence level contours and best-fit values can be computed for individual cosmological parameters α and $\Omega_{\rm m}$.

$$\mathcal{L}_\alpha = \int \mathcal{L}'(\alpha, \Omega_{\rm m}) d\Omega_{\rm m} \text{ and } \mathcal{L}_{\Omega_{\rm m}} = \int \mathcal{L}'(\alpha, \Omega_{\rm m}) d\alpha \ .$$

- We define 1, 2, and 3σ confidence level intervals (α_1, α_2) and $(\Omega_{\rm m1}, \Omega_{\rm m2})$ so that

$$s = \int_{\alpha_1}^{\alpha_2} \mathcal{L}_\alpha d\alpha \Big/ \int_{\text{all } \alpha} \mathcal{L}_\alpha d\alpha \ ,$$

$$s = \int_{\Omega_{\rm m1}}^{\Omega_{\rm m2}} \mathcal{L}_{\Omega_{\rm m}} d\Omega_{\rm m} \Big/ \int_{\text{all } \Omega_{\rm m}} \mathcal{L}_{\Omega_{\rm m}} d\Omega_{\rm m} \ ,$$

and s equals to 68.27, 95.45, and 99.73 respectively.